Right Region, Wrong Reference: Diagnosing Unsupported Grounding in MLLMs

Anonymous ACL submission

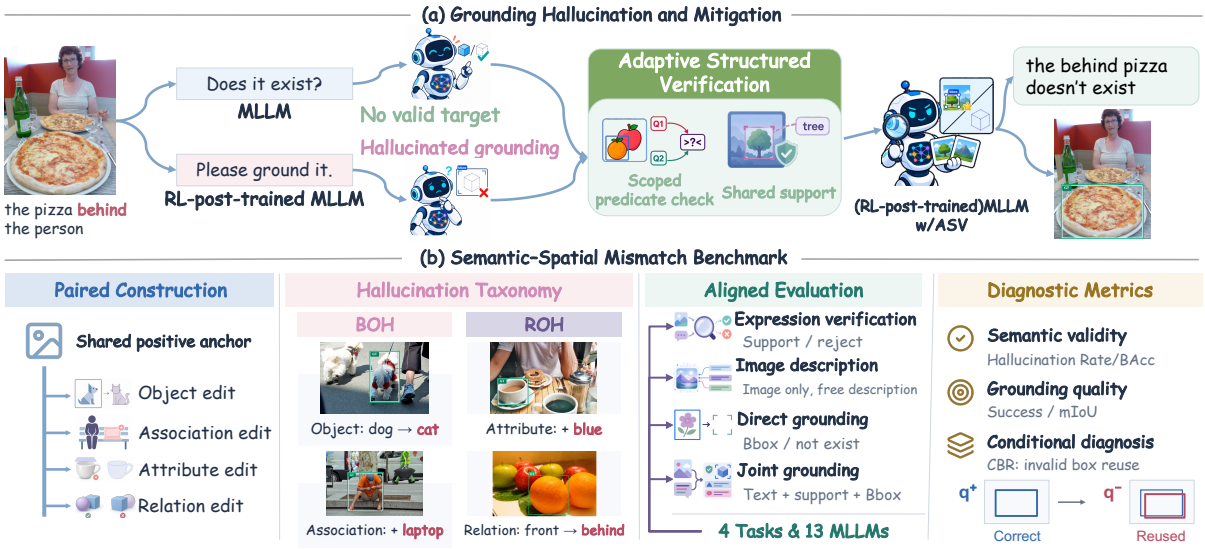


Figure 1: **S²M-Bench overview.** (a) Language rejection can coexist with unsupported grounding; ASV verifies candidate support and scoped predicates. Green boxes mark positive reference regions. (b) Shared positive anchors yield four EOH/ROH edits, evaluated through four aligned tasks and diagnostic metrics, including conditional reuse after correct positive grounding.

Abstract

Multimodal large language models (MLLMs) can erroneously output bounding boxes for absent targets, a failure we term **Semantic–Spatial Mismatch**—semantic support is absent, yet the model still makes a spatial commitment. To systematically investigate this, we distinguish **Existential Object Hallucination (EOH)**, where a required object is absent, from **Referential Object Hallucination (ROH)**, where visible objects fail to satisfy requested attributes or relations. Accordingly, we introduce **S²M-Bench**, a comprehensive evaluation framework, alongside **Conditional Box Reuse (CBR)**, a metric measuring the inappropriate reuse of regions from supported counterparts. Extensive evaluations reveal that current MLLMs are significantly more vulnerable to ROH than EOH, frequently prioritizing visual salience over relational fidelity even when semantic support is lacking. To address this limitation, we propose **Adaptive Structured Verification (ASV)**, an inference-time method that verifies complete-expression support and selectively compares competing predicates. ASV effectively reduces the mean false-grounding rate from 59.23% to 19.16% while largely preserving correct groundings, establishing explicit semantic verification as an indispensable mechanism for reliable spatial localization.

1 Introduction

Visual grounding requires both a valid reference and an accurate region. Yet, current multimodal large language models (MLLMs) frequently hallucinate the latter without the former, assigning plausible bounding boxes to unsupported expressions—such as grounding "the cup on the right" when only a left cup exists. We term this fundamental error **Semantic–Spatial Mismatch**, where spatial commitment persists without underlying semantic support. To achieve reliable grounding, models must therefore establish com-

plete semantic support before committing to any spatial target.

Existing evaluations approach semantic validity and spatial accuracy from different directions. CHAIR (Rohrbach et al., 2018), POPE (Li et al., 2023), and AMBER (Wang et al., 2023) assess claims or judgments without directly evaluating regions. Conventional grounding benchmarks evaluate spatial overlap for expressions with valid targets (Mao et al., 2016). FineCops-Ref (Liu et al., 2024) tests compositional negatives and confidence ranking, while CSR (Li et al., 2026) diagnoses counterfactual errors and KnowDR-REC (Jin et al., 2026) examines invalid binding. We connect these perspectives by testing whether models reject unsupported references yet still assign them regions when the same requests are independently submitted to each queried task.

To characterize this mismatch, we distinguish **Existential Object Hallucination (EOH)**, requiring an absent object, from **Referential Object Hallucination (ROH)**, where visible objects fail to satisfy an attribute or relation. Object and association edits instantiate EOH; attribute and relation edits instantiate ROH. For ROH, locating a visible object is insufficient: the entire expression must be supported by its attributes and relations. This distinction motivates two questions: **1)** *How do MLLMs exhibit EOH and ROH in language judgments and grounding outputs?* **2)** *Does unsupported region reuse persist after a model has correctly grounded the paired positive expression?*

To answer these questions, we introduce **S^{2}M-Bench**, which pairs each supported anchor with four unsupported variants, yielding 4,000 pairs from 1,000 image groups. The shared image and positive anchor hold the visual context fixed while changing the challenged semantic constraint. Expression verification, direct grounding, and joint grounding process the same requests independently, while image captioning is query-free. **Conditional Box Reuse (CBR)** measures whether unsupported expressions receive nearly the same regions as their correctly grounded positive counterparts, isolating reference errors after successful localization.

Evaluating 13 MLLMs on this benchmark, we find similar positive grounding for general-purpose and RL-adapted models but mean direct-grounding false-grounding rates of 37.64% and 68.83%, respectively. ROH is harder to reject than EOH in every model, and CBR reveals invalid region reuse after correct positive grounding. These findings motivate **Adaptive Structured Verification (ASV)**, which checks complete-expression support and selectively compares competing relation predicates before accepting candidates. The support check covers both EOH and ROH; selective predicate comparison strengthens rejection of structural errors. ASV preserves candidate coordinates and reduces equal-model mean false grounding from 59.23% to 19.16%, retaining 91.71% of initially correct groundings on average.

Our contributions are:

- **S^{2}M-Bench.** A benchmark comprising 4,000 positive–negative pairs from 1,000 image groups, with an EOH/ROH taxonomy, four aligned tasks, and CBR for diagnosing semantic–spatial mismatch.
- **Adaptive Structured Verification.** A verification policy derived from benchmark diagnosis that checks complete-expression support and selectively compares relation predicates.
- **Empirical findings.** Evidence across 13 general-purpose and RL-adapted MLLMs that ROH is harder to reject under explicit queries and unsupported region reuse persists after correct positive grounding.

2 Related Work

2.1 Existential Object Hallucination

Existential Object Hallucination (EOH) concerns the absence of required objects. Captioning and presence evaluations measure unsupported object claims (Rohrbach et al., 2018; Li et al., 2023). Decoding interventions and post-generation verification target these errors (Huang et al., 2024; Leng et al., 2024; Yin et al., 2024). In grounding, rejection must also prevent unsupported box predictions. OpenRef uses multi-task consistency checking for this purpose (Wu and Zhang, 2026). Open-world REC includes multi-target and no-target expressions in GREC, and varying target counts in RECANTFormer (He et al., 2023; Hemanthage et al., 2024). Consistent outputs may still affirm unsupported objects. S^{2}M-Bench traces these errors across four tasks by linking expression judgments and descriptions to box emission.

2.2 Referential Object Hallucination

However, object existence does not guarantee the queried attributes or relations, whose violation defines Referential Object Hallucination (ROH). Existing evaluations cover generated text and discriminative answers (Wang et al., 2023; Wada et al., 2026), while grounding must also tie these errors to a specific region. Grounding methods improve region selection through reinforcement learning (Bai et al., 2025c; Liu et al., 2025c) or iterative semantic reasoning (Luo et al., 2026). FineCops-Ref evaluates reference validity with edited negative expressions (Liu et al., 2024). KnowDR-REC separates entity resolution, grounding, and referring validity, and identifies binding hallucination with knowledge-conditioned counterfactuals (Jin et al., 2026). RC-GRPO trains models to reject nonexistent targets while preserving positive grounding (Yang et al., 2026). S^{2}M-Bench links expression judgments to box outputs and applies candidate verification for post-hoc box retention.

3 S^{2}M-Bench: Semantic–Spatial Mismatch

S^{2}M-Bench tests whether MLLMs withdraw unsupported region assignments. Its paired design fixes the image and supported grounding anchor while changing the challenged semantic constraint, linking semantic rejection to region prediction.

3.1 Problem Formulation

Given image I and complete expression q, let $\mathcal{R}(I, q)$ contain the visible instances satisfying every required object, attribute, and relation constraint. We define the support of a complete expression as follows:

$$S(I, q) = \mathbf{1}[\mathcal{R}(I, q) \neq \varnothing]. \quad (1)$$

A candidate $\hat{b}$ selecting instance $z(\hat{b})$ is correctly bound when $z(\hat{b}) \in \mathcal{R}(I, q)$. Geometric accuracy measures overlap with the target annotation b^*. A tight box can select an instance that violates the expression; an imprecise boundary can surround the correct instance. For $S(I, q) = 0$, the correct grounding output is empty.

Semantic scope. The EOH/ROH distinction applies to every object required by the complete expression: a missing reference object belongs to EOH, while ROH requires the referenced objects to be visible. Observable errors include accepting an unsupported expression, stating its unsupported claim, or assigning it a region.

3.2 Controlled Construction and Human Review

Shared positive anchors. Each group retains a RefCOCOg image I_i, supported expression q_i^+, and positive target box b_i^* (Mao et al., 2016). Four unsupported variants on the same image yield 1,000 image groups, 4,000 positive–negative pairs, and 5,000 image–expression items. Existing negatives are reused only when their positive expression matches the anchor; an MLLM supplies missing types. The target box annotates the supported positive query.

Four semantic edits. Object and association edits instantiate EOH. An *object* edit replaces an object mention with an absent category; an *association* edit substitutes or adds a contextually plausible but absent object suggested by the visible scene (Li et al., 2023). Attribute and relation edits instantiate ROH while keeping the referenced objects visible. An *attribute* edit assigns an unsupported property; a *relation* edit changes a predicate or entity roles. Records identify the missing object and textual edit for EOH. ROH records identify the changed property or the subject, relation, and reference object.

Image-wide validation and human review. The `qwen-max` annotator supplies semantic units and risk flags after completing the four edits, with GPT and Gemini cross-checking the generated annotations. Four researchers experienced in MLLM dataset construction review every QA item (100% coverage). Reviewers' image assignments overlap, and each item has one primary reviewer, who inspects the image and generated annotation to check image–text correspondence, positive-query validity, negative-query invalidity, EOH/ROH type, and the positive target region. EOH review verifies required-object absence throughout the image; ROH review verifies the visibility of referenced objects and excludes any valid target satisfying the edited constraint, including redirected references. Structural checks verify one shared positive, four negative types, and aligned semantic units per group. Appendix B.2 details the review criteria; Figure 2 illustrates sample construction and human review.

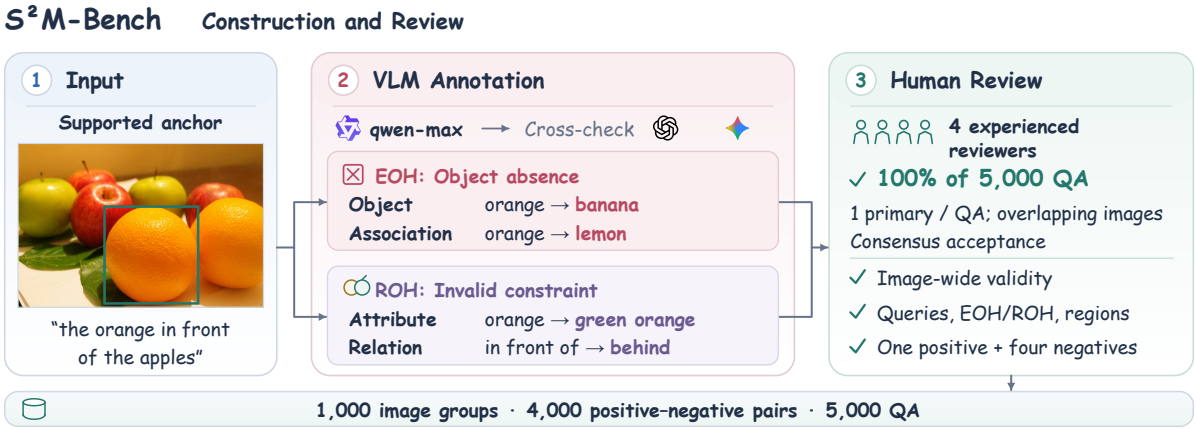


Figure 2: **S²M-Bench: Construction and Review.** A real image and its supported query anchor four controlled semantic edits: Object and Association (EOH), and Attribute and Relation (ROH). The `qwen-max` annotator supplies paired queries and semantic annotations, cross-checked by GPT and Gemini before human review of every QA item with overlapping image assignments. Each reviewed image group contains one positive query and four negative queries.

Task	Input, output, and primary measures
Expression verification	Image and expression → support; false acceptance and balanced accuracy.
Direct grounding	Image and expression → box or empty; FGR, mIoU, and CBR.
Image captioning	Image only → free description; EOH/ROH false assertion, object hallucination, and coverage.
Joint grounding	Image and expression → description, support, and box or empty; FGR, mIoU, CBR, and caption hallucination.

Table 1: Each queried task uses the same 1,000 positive and 4,000 negative image–expression items. Image captioning uses the corresponding images without a query.

3.3 Tasks and Metrics

Aligned outputs. Each queried task in Table 1 processes the same image–expression item independently. Joint grounding returns description, support, and region together; image captioning is query-free.

Acceptance and claims. Verification FAR reports false acceptance among valid support judgments on negative queries; balanced accuracy averages positive acceptance and negative rejection within valid judgments. Caption assertion rate measures false assertion of the annotated negative semantic units in query-free descriptions, split into EOH and ROH (Appendix B.5). Object-level hallucination and target coverage additionally assess description scope (Rohrbach et al., 2018; Wang et al., 2023).

For negatives $\mathcal{D}^-$ ($N^- = |\mathcal{D}^-|$), FGR counts parsed box emissions e_i:

$$\mathrm{FGR} = \frac{1}{N^-} \sum_{i \in \mathcal{D}^-} e_i. \tag{2}$$

Degenerate emissions count with zero IoU. Both grounding tasks use this parsed-emission definition in Table 2 and Appendix C. All-positive mIoU includes empty, rejected, or invalid predictions as zero. Correct grounding requires IoU ≥ 0.5.

Conditional box reuse. CBR measures positive $\hat{b}_i^+$ reuse by type-t negative $\hat{b}_{i,t}^-$:

$$\begin{aligned} c_i &= \mathbf{1}[\mathrm{IoU}(\hat{b}_i^+, b_i^*) \geq \tau], \\ r_{i,t} &= \mathbf{1}[\mathrm{IoU}(\hat{b}_{i,t}^-, \hat{b}_i^+) \geq \rho], \\ \mathrm{CBR}_t &= \frac{\sum_i c_i r_{i,t}}{\sum_i c_i}. \end{aligned} \tag{3}$$

With $\tau = 0.5$ and $\rho = 0.8$, empty or invalid boxes have zero IoU. Each model's task shares denominator $n_c = \sum_i c_i$ across all four types, retaining every paired negative, including refusals. CBR is undefined at $n_c = 0$.

Retained correct grounding. Positive keep decisions k_i^+ determine how many initially correct groundings survive the filtering process:

$$R_{\mathrm{correct}} = \frac{\sum_i c_i k_i^+}{\sum_i c_i}. \tag{4}$$

Filtering fixes original eligibility c_i and positive reference boxes. We report achieved retention; Appendix B.3 specifies output-state and boundary rules.

4 Method

Adaptive Structured Verification (ASV) tests an upstream MLLM's candidate against the complete expression, addressing the mismatch in Section 3.1. Shared candidate support covers all four negative types. A selective predicate comparison supplies additional evidence for rejecting structural errors. Figure 3 presents both stages, which preserve the image and candidate coordinates.

Complementary candidate support. JEV-2B adapts the TypeSafe/Open-Jev interface to score complete-expression support for $(I, q, \hat{b})$ (Almeida, 2026; Cai, 2026). Its scalar head a_ψ reads the final prompt representation h_ϕ in one multimodal forward pass. Temperature scaling (Guo et al., 2017) produces a support logit, whose sigmoid estimates support probability. Frozen OmniVerifier-7B (Zhang et al., 2026) scores the full image marked with $\hat{b}$ using the true-minus-false token-logit difference at the answer position:

$$\begin{aligned} s_{\mathrm{J},0} &= a_\psi(h_\phi(I, q, \hat{b}))/T, \\ s_{\mathrm{O},0} &= \ell_{\mathrm{true}} - \ell_{\mathrm{false}}, \end{aligned} \tag{5}$$

For calibration temperature $T > 0$, larger scores favor support. A logistic head f_0 combines $\mathbf{s}_0 = (s_{\mathrm{O},0}, s_{\mathrm{J},0})$ to verify the complete expression. JEV adaptation and temperature fitting follow the procedure in Appendix D.2, with OmniVerifier and upstream MLLMs remaining frozen.

Selective predicate competition. For a non-empty candidate, $p(q)$ is the longest predicate matched by a frozen inventory. Its opposition scope $\mathcal{P}_{\mathrm{opp}}$ includes *behind*, *in front of*, *under*, *on top of*, *above*, and *below*. Competition requires both original scores, both alternative-score maxima, and a fitted structural head. Writing this availability as a_i, the route is $g_i = \mathbf{1}[p(q_i) \in \mathcal{P}_{\mathrm{opp}}]a_i$.

Matched replacement rules construct $\mathcal{Q}(q)$, preserving object mentions, excluding q, and removing duplicates (Appendix D.3). Each verifier compares alternatives on the same image and box:

$$\delta_v = \max_{q' \in \mathcal{Q}(q)} s_v(I, q', \hat{b}) - s_{v,0}, \quad v \in \{\mathrm{J}, \mathrm{O}\}. \tag{6}$$

A positive margin favors an alternative. These within-verifier contrasts join the original scores as $\mathbf{z} = [\mathbf{s}_0, \delta_{\mathrm{O}}, \delta_{\mathrm{J}}]$, supplying absolute support and relative preference to logistic head f_1.

Monotone decisions and retention. Let e_i indicate an upstream candidate. With available support scores and a fitted head, $k_i^{(0)} = e_i \mathbf{1}[f_0(\mathbf{s}_{0,i}) \geq \eta_0]$; otherwise $k_i^{(0)} = e_i$. The final output follows a monotone rule combining these decisions:

$$k_i = \begin{cases} k_i^{(0)}, & g_i = 0, \\ k_i^{(0)} \mathbf{1}[f_1(\mathbf{z}_i) \geq \eta_1], & g_i = 1. \end{cases} \tag{7}$$

Thus $k_i \leq k_i^{(0)}$: competition preserves all rejections from shared support. Missing structural evidence falls back to shared support. We return the unchanged upstream box when $k_i = 1$ and empty otherwise.

Heads and thresholds use image-level two-fold cross-fitting per upstream MLLM, grouping each image's paired queries. Each training fold fits heads and thresholds; the other fold supplies evaluation outputs. Thresholds target nominal 0.95 retention on originally correct positives, with eligibility fixed before filtering. We report achieved retention after both gates and all-positive mIoU; Appendix D provides calibration details and fallback rules.

5 Experiments and Analysis

We first analyze the unfiltered models on S^{2}M-Bench, then evaluate the verification framework from Section 4 on the same benchmark. All task evaluations use the benchmark's 1,000 image groups. Each queried task receives 1,000 positive and 4,000 negative expressions; query-free captioning produces one description per image. Direct and joint grounding use the parsed-emission FGR in Equation 2; reported differences are computed from the displayed means.

5.1 Benchmark Evaluation

5.1.1 Language–Grounding Mismatch on Matched Queries

We evaluate four general-purpose and nine RL-adapted MLLMs on 1,000 image groups. Table 2 reports verification FAR, caption assertion rates, and spatial FGR across four EOH/ROH tasks, including two backbones w/ ASV. Figure 4 com-

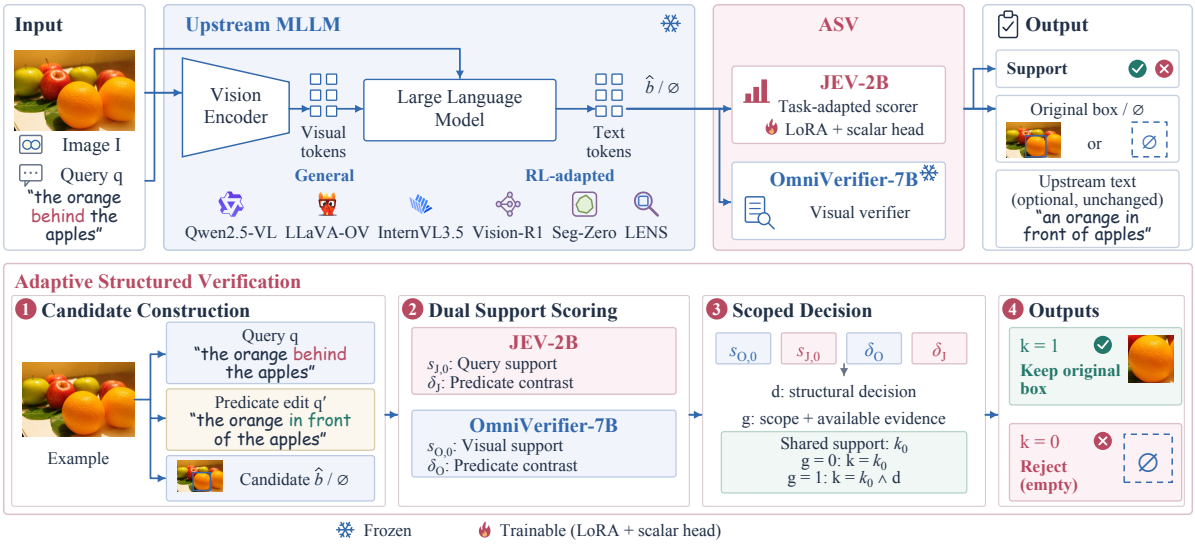


Figure 3: **Adaptive Structured Verification.** Top: frozen upstream MLLMs provide candidates to ASV; optional descriptions pass through unchanged. Bottom: candidate construction, dual support scoring, scoped decisions, and outputs. JEV and frozen OmniVerifier supply shared support and predicate contrasts on the same image and candidate. The monotone gate retains the original box or returns empty (Equation 7). Snowflakes mark frozen components; flames mark JEV's trainable LoRA adapters and scalar head.

Models	EOH				ROH			
	Verif. FAR↓	Caption Assert.↓	Direct-*G* FGR↓	Joint-*G* FGR↓	Verif. FAR↓	Caption Assert.↓	Direct-*G* FGR↓	Joint-*G* FGR↓
General models ($n = 4$)								
Qwen3-VL-8B	**11.6**	10.1	**10.1**	11.9	35.8	3.6	32.2	33.0
Qwen2.5-VL-7B	11.8	8.6	20.0	19.5	**32.9**	3.4	49.0	48.2
InternVL3.5-8B	14.3	8.1	41.6	27.7	38.2	3.1	80.9	52.0
LLaVA-OV-7B	28.8	**4.1**	18.2	90.9	62.1	**1.5**	49.1	95.5
RL models ($n = 9$)								
LENS-3B	25.1	**7.7**	90.4	24.0	49.6	**2.5**	98.0	**33.1**
Orsta-7B	15.1	8.7	27.9	22.6	37.8	3.2	60.4	52.5
Seg-R1-7B	**10.2**	9.7	47.0	17.9	**31.4**	3.5	76.5	45.3
Seg-Zero-7B	29.6	9.2	95.3	23.7	60.8	3.6	98.9	52.3
TreeVGR-7B	48.9	8.9	62.2	42.7	84.9	3.2	89.0	76.3
UniVG-R1-7B	25.4	7.8	99.7	78.7	48.5	2.6	99.9	96.1
Vision-R1-7B	16.8	8.9	22.3	27.9	43.9	3.0	53.3	59.7
VisionReasoner-7B	28.3	8.7	73.5	30.5	56.2	3.5	88.2	61.4
Visual-RFT-7B	13.4	9.0	17.2	56.2	39.7	3.7	39.3	74.9
Models w/ ASV								
InternVL3.5-8B w/ ASV	14.3	8.1	14.8	**11.5**	38.2	3.1	**31.9**	**28.7**
UniVG-R1-7B w/ ASV	25.4	7.8	**12.8**	**15.2**	48.5	2.6	**32.9**	37.5

Table 2: Four-task rates for 13 MLLMs (%). *G* denotes grounding; B denotes billion parameters. FAR denotes false acceptance in verification; Assert. denotes false assertion of annotated negative units in captions (Appendix B.5). FGR counts parsed coordinate emissions in both grounding tasks. ASV preserves upstream language outputs. Best/second-best values are bold/underlined within each family, including its corresponding ASV configurations.

pares verification with grounding on identical negative queries. Means weight models equally; Appendix C adds grounding quality and description coverage.

Semantic rejection can accompany region emission. Across 12 MLLMs, 26.72% of matched negative queries are explicitly rejected in verification yet assigned a region in direct grounding. This mismatch spans all four edit types (22.60–28.47%; Appendix C.8). Gen-

	False grounding (%)			CBR (%) ↓				Positive grounding	
Strategy	FGR ↓	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct} ↑	mIoU ↑
Unfiltered	59.23	48.11	70.36	16.47	25.63	33.96	42.79	**100.00**	**0.4268**
Shared support verification	22.55	12.20	32.89	**1.69**	9.22	8.32	27.14	94.26	0.4011
ASV	**19.16**	**11.94**	**26.38**	**1.69**	**9.07**	**8.19**	**19.34**	91.71	0.3910

Table 3: Equal-model mitigation across 13 MLLMs at calibration target 0.95. R_{correct} is achieved correct-grounding retention. CBR fixes original eligibility and positive boxes. Rates are percentages; mIoU covers all positives. Best values are bold.

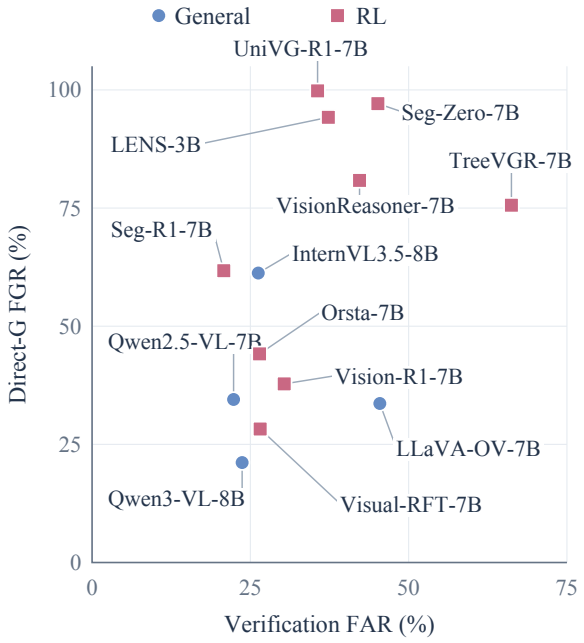


Figure 4: **Language acceptance and region emission.** Verification FAR (horizontal) and direct-grounding FGR (vertical) on negative queries. Each point is one MLLM; circles/squares denote General/RL families.

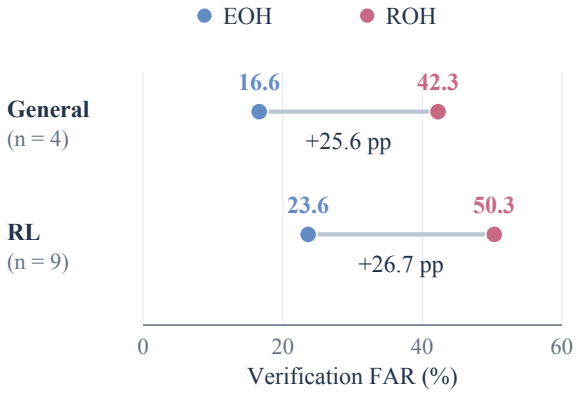


Figure 5: **EOH/ROH difficulty in verification.** Equal-model verification FAR for General ($n = 4$) and RL ($n = 9$) MLLMs; labels along the connectors show ROH-minus-EOH gaps (percentage points).

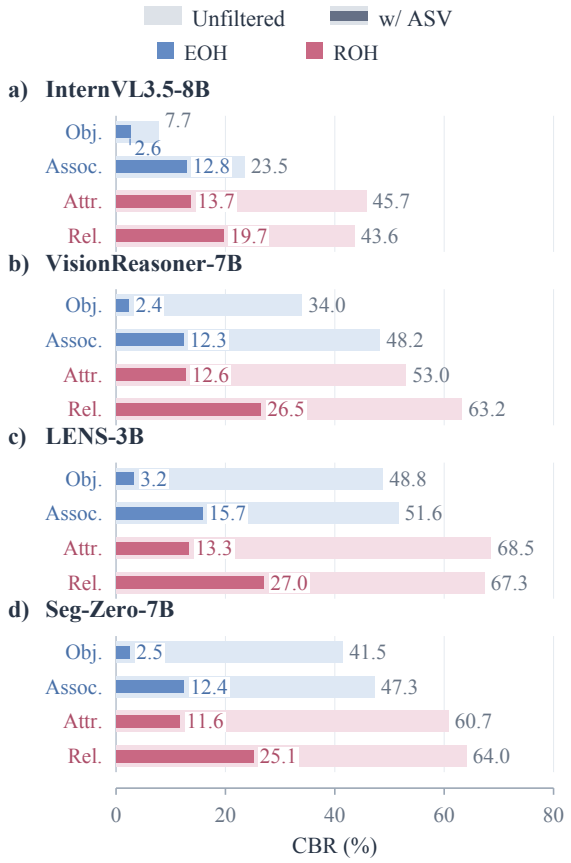


Figure 6: **Direct-grounding mitigation across four MLLMs.** Light and dark inset bars show unfiltered and model w/ ASV CBR, using the same 0–80% scale, fixed candidates, and original positive eligibility.

eral and RL MLLMs have similar positive mIoU (0.4331/0.4240) and grounding success (45.0/42.9%). Their verification FAR differs by 7.30 points (29.44/36.74%), whereas FGR differs by 31.19 points (37.64/68.83%), showing why grounding warrants separate evaluation.

Joint grounding lowers FGR in nine MLLMs and increases it in four; all-positive mIoU is lower in twelve MLLMs (Appendix C.7).

		Direct grounding		Joint grounding	
Family	Models	EOH	ROH	EOH	ROH
General	4	11.24	34.26	23.45	40.91
RL	9	25.41	40.20	14.08	31.20

Table 4: Family CBR (%), extending the paired-grounding analysis of KnowDR-REC (Jin et al., 2026): equal-model means on original positive-correct sets, averaging two edit types per hallucination level.

5.1.2 ROH Is Harder to Reject under Explicit Queries

Every evaluated MLLM has higher verification FAR on ROH than EOH, a 26.35-point mean gap. Figure 5 shows EOH/ROH rates of 16.63/42.25% for general-purpose MLLMs and 23.64/50.31% for the nine RL-adapted MLLMs.

Direct-grounding EOH/ROH FGR is 22.47/52.80% for General and 59.50/78.17% for RL. ROH exceeds EOH in all 13 MLLMs (means: 70.36% versus 48.11%; gap: 22.25 points; Appendix Table A5). Joint-grounding FGR is higher for ROH than EOH in all 13 MLLMs (Table 2).

Assertions in query-free generation. Captions assert EOH negative units more frequently than ROH units in all 13 MLLMs: the rates are 7.73/2.90% for General and 8.73/3.20% for RL. These rates characterize spontaneous semantic assertions; explicit-query rejection is evaluated by verification and grounding.

5.1.3 Conditional Reuse Reveals Referential Errors

Building on KnowDR-REC's positive-correct-conditioned Neg-Pos audit (Jin et al., 2026), we extend box-reuse analysis to EOH/ROH, four controlled semantic edits, and both grounding tasks. CBR fixes each MLLM's positive-correct set and retains every paired negative in the denominator, including empty outputs (Appendix Table A6). The four MLLMs with the most successful direct-grounding positives reuse their regions on 55.12–64.00% of relation negatives after correctly grounding the positive target.

ROH reuse exceeds EOH in both families and tasks (Table 4), with direct-grounding gaps of 23.02/14.79 points for General and RL. All 13 MLLMs exhibit higher direct-grounding ROH than EOH CBR (Appendix Table A6) on each model's original positive-correct set.

On its original positive-correct set, Qwen2.5-VL's object/relation CBR is 4.55/44.63%. Relation reuse exceeds object reuse in all 13 MLLMs under both direct and joint grounding; VisionReasoner's joint relation reuse remains 55.1%. These errors persist on correctly grounded positive anchors.

5.2 Intervention Results

Table 3 reports equal-model operating points for the same 13 MLLMs at nominal target 0.95, with fixed candidates, original positive eligibility, and mIoU over all positive queries.

5.2.1 The Diagnosed Assignments Can Be Reduced

All four types improve over unfiltered predictions. FGR falls from 59.23% to 19.16%, a 40.07-point reduction (67.65% relative), with lower CBR for all four types. Correct-grounding retention is 91.71%; mIoU changes from 0.4268 to 0.3910. ASV lowers FGR in all 13 MLLMs under both grounding protocols relative to unfiltered predictions and shared support (Appendix D; Table A17). Figure 6 shows four representative MLLM mitigation profiles.

Joint grounding also benefits. Its separate configuration uses joint candidates, two support scores, and OmniVerifier's predicate contrast (Appendix D.9; Table A16). All four mean CBR values decrease from unfiltered predictions, with 92.99% retention and mIoU changing from 0.3639 to 0.3341. Beyond shared support, relation CBR decreases from 25.18% to 23.08%, without increasing any of the four means.

5.2.2 Structural Comparison Strengthens ROH Rejection

Shared support reduces EOH FGR from 48.11% to 12.20% and ROH from 70.36% to 32.89%. Structural comparison further lowers ROH FGR to 26.38%, a 6.51-point reduction, while EOH reaches 11.94%. Relation CBR falls from 27.14% to 19.34%; object CBR remains 1.69% at reported precision. Retention changes from 94.26% to 91.71%, and mIoU from 0.4011 to 0.3910. At matched achieved retention across 12 MLLMs, interpolated full-policy rates lower relation FGR and CBR by 7.54 and 5.58 percentage points relative to shared support (paired model-bootstrap 95% CIs: [5.99,9.03] and [4.05,7.05]; Appendix D.6).

Serial structural calibration (Appendix D.6.1)

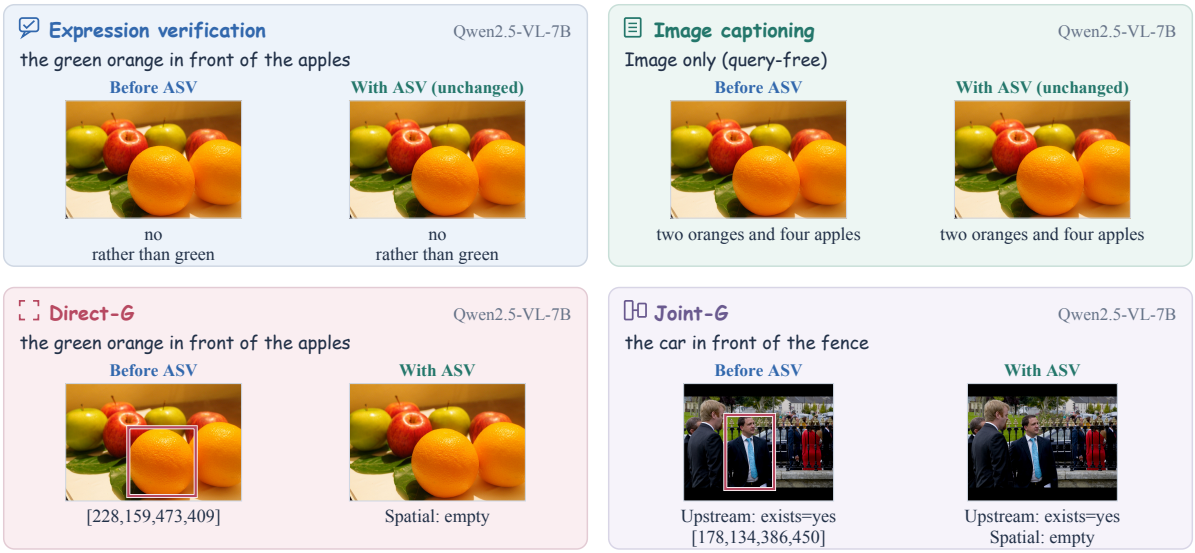


Figure 7: **Representative outputs before and with ASV.** Qwen2.5-VL rejects the unsupported attribute in verification yet emits a region in Direct-*G*. ASV rejects the illustrated Direct-*G* and Joint-*G* candidates while preserving upstream text. Captioning shows the query-free output on the same image. The illustrated Direct-*G* response uses an auxiliary attention-inspection prompt and is excluded from aggregate evaluation.

also attains FGR 20.13% versus 22.55% for shared support, with correct-grounding retention of 92.49% versus 94.26% across the same 13 evaluated upstream MLLMs.

Image-swap probes give a 13-model mean OmniVerifier logit shift of 11.41 (Appendix D.7), supporting visual evidence in expression verification.

Figure 7 illustrates the task-level outputs: ASV removes unsupported spatial candidates while retaining the upstream language responses. Appendix D.10 specifies the attention visualization protocol; case selection is pending.

6 Conclusion

S^2M-Bench examines unsupported region assignments by MLLMs. Its EOH/ROH taxonomy and aligned benchmark expose semantic errors that positive grounding alone misses; CBR identifies invalid reuse after successful positive grounding. Adaptive Structured Verification reduces false assignments across four negative types while retaining initially correct groundings, supporting joint assessment of geometry and validity of the complete reference.

Limitations

Our evaluation covers 1,000 RefCOCOg image groups and 13 MLLMs across four semantic edit types. The reported family summaries characterize these evaluated systems, and CBR conditions on each model's correctly grounded positives. Verifier adaptation uses a separate image pool with the same semantic construction scheme.

Adaptive Structured Verification filters fixed upstream candidates using complete-query support and a frozen predicate inventory. Its operating points are defined by image-level two-fold cross-fitting and the measured retention of originally correct positives. This evaluation scope focuses on the semantic validity of candidate assignments and preserves upstream box coordinates.

Ethics Statement

The study uses existing research images and controlled unsupported expressions as evaluation cases. Human review validates the image–expression annotations before evaluation. Candidate filtering reduces unsupported visual claims while tracking correct-grounding retention. Source images, derived annotations, and model predictions are distinguished in the evaluation records.

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A Semantic Validity in Visual Grounding

The distinction between accurate localization and valid reference motivates the paired evaluation developed below. Appendix A clarifies this distinction and its empirical basis; Appendix B specifies benchmark construction and scoring; Appendix C provides the complete unfiltered results; and Appendix D details the mitigation settings and ablations.

A.1 Localization and Referential Validity

Accurate localization alone does not establish that a model understood the referring expression. A box establishes geometry, whereas a valid reference requires an instance satisfying the complete expression. A model can locate a visible cup accurately and still assign that cup to an unsupported attribute or relation. With the notation of Section 3.1, geometric overlap with b^* does not imply $z(\hat{b}) \in \mathcal{R}(I, q)$. For an unsupported expression, $\mathcal{R}(I, q) = \varnothing$, so any returned region is an invalid assignment. This motivates reporting positive localization and negative-query emission separately. Conditional box reuse (CBR) examines whether the invalid assignment persists in image groups whose paired positive was correctly grounded. It is a behavioral diagnostic, not a claim about the model's internal reasoning.

A.2 Relation to Existing Evaluation

Existing hallucination and grounding benchmarks address complementary aspects of semantic validity. RefCOCOg supplies supported referring expressions and target regions (Mao et al., 2016). CHAIR evaluates hallucinated object mentions in captions, POPE probes object existence, and AMBER covers several hallucination dimensions (Rohrbach et al., 2018; Li et al., 2023; Wang et al., 2023). FineCops-Ref already studies compositional negatives and rejection; CSR and KnowDR-REC further examine counterfactual segmentation and knowledge-conditioned grounding (Liu et al., 2024; Li et al., 2026; Jin et al., 2026). KnowDR-REC also reports a positive-correct-conditioned negative–positive overlap audit, which is a direct antecedent of this style of conditional diagnosis. Thus, unsupported references, rejection, and conditional overlap are established concerns.

Our diagnostic contribution is to hold the image and supported anchor fixed across four edited negatives and compare independently submitted verification and grounding requests, with query-free captions as a separate view. EOH concerns a required but absent object; ROH concerns visible objects that fail the complete attribute or relation constraint. CBR conditions this comparison on successful positive grounding. Complete per-model results are reported once in Appendix C; their role is to test this distinction rather than redefine earlier benchmarks.

A.3 Empirical Basis for the Diagnosis

Three observations motivate examining invalid assignments separately from failures of positive localization or differences between unrelated examples. First, the general-purpose and RL-adapted groups have similar positive mIoU (0.4331/0.4240) but different direct-grounding FGR (37.64/68.83%). Second, the query-matched analysis reports explicit verification rejection alongside direct-grounding emission on 26.72% of matched negative records across 12 models (Appendix C.8). Third, relation CBR exceeds object CBR in every evaluated model, even after conditioning on correct positive grounding (Appendix C.6).

These findings distinguish successful localization from valid assignment. They do not identify a causal effect of RL: the model families differ in backbones, supervision, and interfaces. Nor does matching independently generated responses establish a contradiction within a single response. Appendix B.5 separates these events, and Appendix C.8 explains why aggregate rate gaps alone cannot count query-level mismatches.

B S²M-Bench: Construction and Evaluation Protocols

We construct unsupported expressions from shared positive anchors and validate them against the full image. The evaluation then distinguishes task-specific output states and fixes the metric denominators, so that annotation validity and model behavior can be assessed separately.

B.1 Shared Anchors and Semantic Edits

To control the image and target when comparing existential and referential errors, each of the 1,000 image groups retains a RefCOCOg image, one supported expression, and its target box (Mao et al., 2016). Four edited negatives share this anchor, giving 4,000 positive–negative pairs and

5,000 distinct image–expression items. The positive item is shared rather than counted as four distinct positives. Existing negatives are reused only when their positive expression matches the anchor; an MLLM supplies missing types.

Object edits replace an object mention with an absent category; association edits introduce a plausible but absent required object. Both are EOH, including when the missing object is the reference object in a relation. Attribute and relation edits preserve the required objects' visibility but invalidate a property, predicate, or participant binding, and instantiate ROH. Association is a construction category, not a claim that co-occurrence proves visual presence; POPE provides related motivation for examining object-existence errors (Li et al., 2023).

B.2 Image-Wide Validation and Human Review

Validation covers the full image to exclude alternative targets that might satisfy an edited expression. EOH review checks that the required object is absent throughout the image. ROH review checks that the required objects are visible and that no alternative target satisfies the edited expression. Relation checks include reference identity, same-class alternatives, exchanged participant roles, and depth or direction ambiguity. The target box annotates only the supported positive.

The `qwen-max` annotator supplies semantic units and risk flags after the four edits; GPT and Gemini cross-check the generated annotations. Four researchers experienced in MLLM dataset construction review all 5,000 items. Each item has one primary reviewer; image assignments overlap, so multiple reviewers can inspect different items from one image. Review covers image–text correspondence, positive validity, negative invalidity, EOH/ROH type, and the target region. Annotations are accepted after reviewers reach agreement. Structural checks enforce one shared positive and all four aligned negative types per group.

Full review coverage does not imply independent duplicate annotation. We therefore distinguish coverage from inter-annotator agreement and do not infer an agreement coefficient from the review protocol.

B.3 Output Parsing and Coordinate Conventions

Scoring distinguishes malformed outputs and refusals from valid geometric predictions. Coordinate conversion is model-specific and precedes clipping and overlap. UniVG-R1 and Visual-RFT emit normalized coordinates in $[0, 1000]$; these are scaled by the actual image width and height. Parsing requires a recognized output structure. Emission, valid geometry, semantic decisions, format failures, and missing executions are distinct fields (Table A1). FGR counts parsed box emissions over all negative queries, including degenerate boxes as emissions with zero IoU. Empty, rejected, or invalid positive predictions contribute zero to all-positive mIoU. A refusal accompanied by a parsed box still counts as an emission. Missing executions require recovery for completeness and must not be interpreted as successful refusals.

Output state	Scoring treatment
Parsed, nondegenerate box	Emission $e = 1$; score geometric IoU.
Parsed box with invalid/degenerate geometry	Emission $e = 1$; IoU and reuse are zero.
Explicit refusal with no parsed box	Emission $e = 0$; zero positive IoU and reuse.
Refusal text plus a parsed box	Record both signals; $e = 1$ and check their contradiction.
Malformed text with no parsed box	Record format failure separately from refusal; retain the query in grounding denominators.
Missing execution or response	Record missing execution or response separately; completeness requires recovered outputs.

Table A1: Grounding-output scoring rules.

B.4 Metric Denominators and Aggregation

Fixed denominators preserve the evaluated population when predictions are filtered. Each model has 1,000 positive queries and 4,000 negatives: 1,000 per edit type and 2,000 per EOH/ROH stratum. CBR uses the originally correct positive set at IoU ≥ 0.5 and tests negative-to-positive box overlap at IoU ≥ 0.8 (Equation 3). The four types share the same denominator within a model and task. Every paired negative, including a refusal, stays in that denominator. Empty or invalid boxes have zero IoU; CBR is undefined if no positive qualifies.

Direct and joint grounding each use their own unfiltered positive-correct set. Within a miti-

gation comparison, both eligibility and positive reference predictions are frozen before filtering. Equal-model means average each model's ratio, rather than pooling eligible groups across models. Correct-grounding retention divides retained originally correct positives by all originally correct positives; it is not the fraction of all positive queries receiving a box. All-positive mIoU and full-denominator FGR accompany retention to expose the cost of rejection.

Verification FAR uses only valid negative support judgments, while balanced accuracy averages positive acceptance and negative rejection within valid judgments. These denominators differ from grounding's full negative-query denominator. The current tables do not report every model's abstention and parse-failure counts, so rate comparisons must retain this qualification.

B.5 Caption Assertions and Cross-Task Alignment

The caption generator receives only the image, without the challenged expression. Under the manuscript's semantic protocol, each model's 1,000 descriptions are checked against four annotated negative units per image, yielding 2,000 EOH and 2,000 ROH judgments. A separate Qwen3.5-9B text-entailment evaluator (Qwen Team, 2026) judges whether the caption asserts each unit. Its fixed deterministic protocol agrees with 63/64 predefined semantic controls (98.44%). This control-set agreement does not establish benchmark-wide evaluator accuracy.

Object mentions are scored with entity binding, attributes with their target entity, and relations with their predicate and participant roles. Affirmative paraphrases count; negations, quotations, and mere lexical co-occurrence do not. Semantic assertion rates differ from the object-level hallucination and target-coverage measures retained in Table A4. CHAIR and AMBER motivate object/claim-level evaluation (Rohrbach et al., 2018; Wang et al., 2023); the study's annotated-unit entailment protocol is a separate measure.

Cross-task records. Records are matched by model and query identifier to compare verification judgments with grounding outputs for the same expression. Explicit support, unsupported judgments, and abstentions remain separate when compared with region emission. A verification rejection paired with direct-grounding emission compares independent task responses. Within joint grounding, an explicit unsupported judgment accompanied by a box is instead a contradiction within one response. These are distinct events; neither should be estimated from an aggregate rate difference.

B.6 Statistical Scope and Sensitivity

The scope of uncertainty depends on the resampling unit. The 13-model mitigation tables report point estimates. For image-group resampling, the specified sampling unit is the source image ID, grouping all model outputs, the positive and four negatives, fold membership, and policy decisions. Numerators and denominators are recomputed before averaging paired changes. With trained strategies fixed, this targets conditional image-sampling uncertainty, not retraining variability.

The matched-retention analysis instead reports paired model-bootstrap intervals from 10,000 resamples over 12 models (Appendix D.6). These intervals characterize variation over that model set, not over images. The manuscript mentions possible CBR sensitivity checks at reuse thresholds $\rho \in \{0.7, 0.8, 0.9\}$ and positive IoU threshold $\tau = 0.75$, but supplies no corresponding numerical table; robustness to those thresholds is therefore not claimed here.

C Extended Evaluation and Analysis

This section reports the complete unfiltered evaluation and examines semantic difficulty, conditional region reuse, and changes induced by the task interface. We first describe the evaluated models and the scope of their outputs, then relate these factors to the observed semantic–spatial mismatch. Mitigation settings and results appear in Appendix D.

C.1 Prompts and Response Formats

Message structure. We separate the system message from the task instruction in the user message. The evaluator places the image and task instruction together in the user message; the expression is supplied for verification, direct grounding, and joint grounding, but omitted for query-free captioning. Below we reproduce the Qwen2.5-VL templates, with `{expr}` denoting the substituted expression. Table A2 records the model-specific system-message handling. The accompanying source supplement retains the full model-specific task templates.

Model	System handling
Qwen2.5-VL, Qwen3-VL, LLaVA-OV, TreeVGR, Orsta (OneRL), Vision-R1	S1
LENS, UniVG-R1	S2
Visual-RFT, Seg-R1, Seg-Zero, VisionReasoner	No explicit system message
InternVL3.5	Separate chat adapter[a]

Table A2: System-message handling in the evaluator. S1 and S2 are reproduced below. [a]The adapter passes the image and user instruction to the model's `chat` method; it does not pass the configured S1 string. The model-internal system text is not established by this adapter.

System S1.

> You are a careful visual grounding assistant.
> Use only visible image evidence. Follow the requested output format exactly.

System S2.

> You are a helpful assistant.

Verification: user text.

> Task: Decide whether the referring expression strictly matches a visible target in the image.
> Referring expression: "`{expr}`"
> The object identity, number, attributes, colors, and spatial relations must all match.
> Answer with exactly one word: yes or no.

Direct grounding: user text.

> Task: Decide whether the referring expression strictly matches a visible target in the image, then localize it if it exists.
> Referring expression: "`{expr}`"
> If the target exists, return exactly one bounding box in absolute image pixels as [x1, y1, x2, y2].
> If the target does not exist, return exactly: not found.
> Do not output explanations.

Query-free captioning: user text.

> Describe this image in two or three sentences.
> Mention only what is clearly visible. Do not guess or infer.
> Output the description only.

Joint grounding: user text.

> Task: First describe the image in one concise sentence. Then check whether "`{expr}`" strictly matches a visible target.
> Step 1 — Describe the image concisely.
> Step 2 — Is there "`{expr}`" in the image?
> The object identity, number, attributes, colors, and spatial relations must all match.
> If yes, provide the bounding box in absolute image pixels as [x1, y1, x2, y2].
> If no, output "not found".
> Return JSON only:



(a) COCO 525342



(b) COCO 560009

Figure A1: Images for the saved responses in Appendix C.2. Verification, direct grounding, and captioning use (a); the joint-grounding example uses (b).

```
{"description":"one concise
sentence", "exists":"yes",
"bbox":[x1,y1,x2,y2]}
{"description":"one concise
sentence", "exists":"no",
"bbox":"not found"}
Do not output explanations.
```

Model-specific formats. The full templates are not identical across models. The LENS, Seg-Zero, VisionReasoner, TreeVGR, and UniVG-R1 interfaces request task-specific `<think>` and/or `<answer>` wrappers; Visual-RFT uses a `Question:` form. For captioning, the standard template above is used for the standard family and as an explicit UniVG-R1 override. The other reasoning-style caption templates request a final description in `<answer>` tags. These are output-format instructions in the user message, not additional system messages. This protocol variation is another reason to interpret model-family comparisons descriptively.

C.2 Representative Evaluation Examples

The following Qwen2.5-VL examples are verbatim saved responses, with line wrapping adjusted for display. They include a correct rejection as well as an unsupported spatial assignment; they are qualitative examples rather than an estimate of error frequency. The first three use COCO image 525342. The joint-grounding example uses image 560009 and therefore is not a fourth response to the same image.

Verification example. Input expression: the green orange in front of the apples.
Saved response: `no`.

Direct-grounding example. Input expression: the green orange in front of the apples.
Saved response: `not found.` The final period belongs to the raw response and is accepted by the refusal parser.

Query-free caption example. **Input:** the image and the captioning instruction above, with no referring expression.
Saved response:

> The image shows a close-up of several fruits, including two oranges and four apples, arranged on a white surface with green leaves partially visible beneath them. The lighting highlights the textures of the fruits' skin.

Joint-grounding example. **Input expression:** the car in front of the fence.
Saved response:

```
“‘json
{"description":"Two men in suits
are conversing near a decorative
fence with people in the
background.", "exists":"yes",
"bbox_2d":[178,134,386,450]}
“‘
```

The saved annotation marks the expression unsupported. Although the description mentions men and a fence, the response asserts existence and emits a region for the car query. The `bbox_2d` key and Markdown fence are preserved from the response; they differ from the requested `bbox` format. ASV acts on the spatial candidate and does not rewrite this upstream description or existence string.

C.3 Evaluated Models and Comparison Scope

The evaluated set comprises four general-purpose and nine RL-adapted MLLMs, as grouped in the manuscript. The descriptions below introduce each model and cite its original publication. The models differ in backbones, task mixtures, supervision, and output interfaces. InternVL3.5 itself incorporates RL (Wang et al., 2025); the general-purpose label does not mean RL-free. Family means are descriptive comparisons over this selected set, not controlled estimates of an RL effect. The citations identify the upstream model families; they do not identify the evaluated checkpoint or isolate any one training factor.

Qwen3-VL-8B (Bai et al., 2025a) Qwen3-VL-8B is an 8B multimodal language model from the Qwen3-VL family. It combines image understanding with instruction following, visual question answering, and language-based reasoning over visual content.

Qwen2.5-VL-7B (Bai et al., 2025b) Qwen2.5-VL-7B is a 7B model from the Qwen2.5-VL family. It supports image understanding, visual question answering, and multimodal instruction following with language responses grounded in the image.

InternVL3.5-8B (Wang et al., 2025) InternVL3.5-8B is an 8B model from the InternVL3.5 multimodal language model family. The family targets general visual perception and multimodal reasoning, including image interpretation and language-conditioned understanding.

LLaVA-OV-7B (Li et al., 2025) LLaVA-OV-7B is a 7B model from the LLaVA-OneVision family. It is intended for general image understanding and multimodal instruction following, with language outputs conditioned on visual and textual inputs.

LENS-3B (Zhu et al., 2026) LENS-3B is a 3B multimodal model with an RL-adapted visual grounding configuration. The model belongs to the class of compact multimodal systems adapted to reason about referring expressions and select corresponding regions.

Orsta-7B (Ma et al., 2025) Orsta-7B is a 7B multimodal model with an RL-adapted visual grounding configuration. It is designed for language-guided visual reasoning in which an expression is associated with one or more candidate image regions.

Seg-R1-7B (You and Wu, 2025) Seg-R1-7B is a model for reasoning-guided visual segmentation and grounding. It combines multimodal reasoning with region-level prediction, supporting visual queries that require identifying and localizing the described content.

Seg-Zero-7B (Liu et al., 2025a) Seg-Zero is a reasoning-chain guided segmentation model trained with cognitive reinforcement. Its reasoning module is trained to guide segmentation decisions, connecting intermediate visual reasoning with the selection of image regions.

TreeVGR-7B (Wang et al., 2026) TreeVGR-7B is a 7B multimodal model with an RL-adapted visual grounding configuration. It represents referring expressions with a reasoning-oriented grounding process that maps language descriptions to visual regions.

UniVG-R1-7B (Bai et al., 2025c) UniVG-R1 is a reasoning-guided universal visual grounding model trained with reinforcement learning. The framework uses supervised reasoning traces and reinforcement learning to support grounding across different visual referring tasks.

Vision-R1-7B (Huang et al., 2025) Vision-R1-7B is a multimodal model with an RL-adapted vi-

sual reasoning and grounding configuration. It is intended to combine explicit visual reasoning with language-conditioned region for image referring queries.

VisionReasoner-7B (Liu et al., 2025b) VisionReasoner-7B is developed for visual reasoning and grounding. The model emphasizes reasoning over visual scenes before producing language responses or spatial assignments for the requested content.

Visual-RFT-7B (Liu et al., 2025c) It is a visual reinforcement fine-tuning approach for visual perception and grounding. It uses verifiable rewards, including localization quality, to improve the connection between perception and predicted regions.

Measure	General-purpose	RL-adapted
Models	4	9
False acceptance (%)	29.44	36.74
FGR (%)	37.64	68.83
Positive success (%)	45.0	42.9
mIoU	0.4331	0.4240
Caption error (%)	3.80	4.51
Target coverage (%)	50.70	57.89
Joint FGR (%)	47.34	48.66
Joint success (%)	33.75	36.42
Joint mIoU	0.3551	0.3679

Table A3: Equal-model family means. Caption metrics evaluate query-free object claims.

General-purpose and RL-adapted models have similar positive grounding success (45.0/42.9%) and mIoU (0.4331/0.4240), while direct FGR differs by 31.19 percentage points (37.64/68.83%). Figure A2 shows the per-model relationship without collapsing the two cohorts' denominators.

C.4 Caption Scope and Explicit Rejection

The scope of a caption differs from that of an explicit grounding request. Query-free captions select their own content; verification and grounding must respond to a supplied expression. In Table A4, object hallucination ranges from 2.00–5.20%, while target coverage ranges from 29.00–62.60%. A caption can omit a difficult target or constraint. Its low object error therefore does not establish that the model will reject an independently submitted false request. Semantic caption assertion rates, reported by EOH/ROH in Table 2, additionally test the annotated negative units under Appendix B.5.

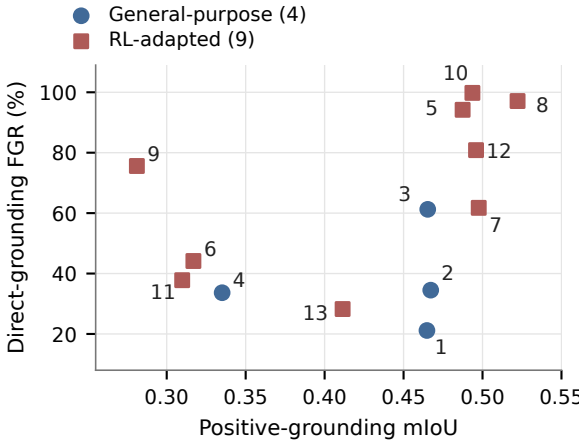


Figure A2: Unfiltered localization and false grounding. Each point is a model, numbered in the order of the model descriptions in Appendix C.3. mIoU uses 1,000 positives and FGR uses 4,000 negatives per model.

MLLM	BAcc	FAR	Obj. err.	Coverage
Qwen3-VL	82.85	23.70	4.40	57.00
Qwen2.5-VL	80.93	22.35	4.40	61.40
InternVL3.5	79.48	26.25	4.40	55.40
LLaVA-OV	75.78	45.45	2.00	29.00
LENS	70.82	37.32	4.20	53.60
Orsta	80.78	26.45	4.80	60.20
Seg-R1	78.60	20.80	5.00	61.60
Seg-Zero	76.32	45.14	5.00	57.00
TreeVGR	66.78	66.23	3.80	49.00
UniVG-R1	73.08	35.62	4.20	60.00
Vision-R1	81.13	30.35	5.20	61.60
VisionReasoner	76.87	42.24	4.60	62.60
Visual-RFT	80.63	26.55	3.80	55.40

Table A4: Expression verification and query-free captioning (%). BAcc/FAR use valid support judgments; object error and target coverage score query-free descriptions. These caption metrics differ from semantic-unit assertion rates.

C.5 Semantic Difficulty across Models

The per-model breakdown in Table A5 shows that the EOH/ROH difference persists within individual models. Verification FAR and direct-grounding FGR are higher for ROH than EOH in all 13 models. The equal-model direct FGR is 48.11/70.36% for EOH/ROH. This supports the paper's within-model claim under explicit verification and direct-grounding queries; it is not a claim about every output interface. A small EOH/ROH gap can coexist with almost universal emission: UniVG-R1 has 99.70/99.90% negative emission, alongside 50.8% positive success. The gap alone is therefore insufficient; the absolute rates and positive performance must also be reported.

MLLM	Verify EOH	Verify ROH	Ground EOH	Ground ROH
Qwen3-VL	11.60	35.80	10.10	32.20
Qwen2.5-VL	11.80	32.90	20.00	49.00
InternVL3.5	14.30	38.20	41.60	80.90
LLaVA-OV	28.80	62.10	18.20	49.10
LENS	25.08	49.60	90.40	98.00
Orsta	15.10	37.80	27.90	60.40
Seg-R1	10.20	31.40	47.00	76.50
Seg-Zero	29.59	60.78	95.30	98.90
TreeVGR	48.95	84.91	62.20	89.00
UniVG-R1	25.35	48.50	99.70	99.90
Vision-R1	16.80	43.90	22.30	53.30
VisionReasoner	28.30	56.18	73.50	88.20
Visual-RFT	13.40	39.70	17.20	39.30

Table A5: False acceptance and direct-grounding FGR (%). Each semantic stratum has 2,000 negative items; verification rates use valid support judgments. Direct-grounding strata cover all 13 evaluated upstream MLLMs.

C.6 Conditional Box Reuse

Conditioning on correct positive grounding allows us to examine false assignments in groups where the model can locate the supported target. Direct-grounding CBR conditions on correctly grounded positives and still shows greater relation than object reuse in all 13 models (Table A6). Each row's four rates share the same eligible positives, including every paired negative refusal. This isolates invalid assignment in groups with successful positive localization, while retaining the conditional scope of the claim. Figure A3 visualizes the four-type profile: relation reuse exceeds object reuse within every model, while absolute rates vary substantially across models. The common color scale supports comparison with the mitigation heatmaps in Appendix D. Joint CBR uses each model's joint positive-correct set. Visual-RFT's object CBR is 3.79% in direct grounding and 17.39% in joint grounding; relation CBR is 37.91/49.76%. UniVG-R1's relation CBR is 55.12/55.06%, while object CBR changes from 38.58% to 25.91%. Because eligibility can change across tasks, these are task-specific conditional profiles, not paired causal changes on an identical successful-positive cohort.

C.7 Effects of the Joint Output Protocol

Requesting a description and support judgment alongside a box changes both rejection and localization behavior. Joint grounding lowers FGR in nine models and increases it in four; all-positive mIoU falls in twelve models. Tables A8 and A9 retain the full positive and negative re-

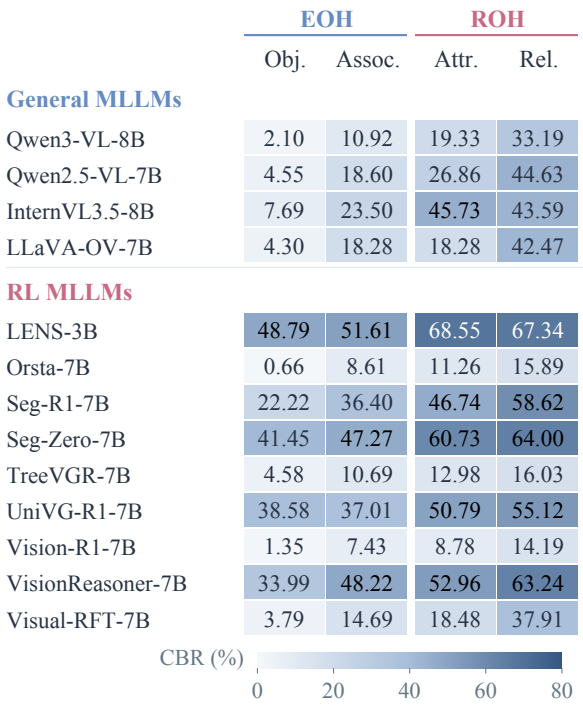


Figure A3: Four-type direct-grounding CBR across all 13 models. Cells show percentages on each model's fixed positive-correct set; lower is better. The four rates in each row share eligibility, while eligibility differs across models (Table A9).

MLLM	Obj.	Assoc.	Attr.	Rel.
Qwen3-VL	2.10	10.92	19.33	33.19
Qwen2.5-VL	4.55	18.60	26.86	44.63
InternVL3.5	7.69	23.50	45.73	43.59
LLaVA-OV	4.30	18.28	18.28	42.47
LENS	48.79	51.61	68.55	67.34
Orsta	0.66	8.61	11.26	15.89
Seg-R1	22.22	36.40	46.74	58.62
Seg-Zero	41.45	47.27	60.73	64.00
TreeVGR	4.58	10.69	12.98	16.03
UniVG-R1	38.58	37.01	50.79	55.12
Vision-R1	1.35	7.43	8.78	14.19
VisionReasoner	33.99	48.22	52.96	63.24
Visual-RFT	3.79	14.69	18.48	37.91

Table A6: Direct-grounding CBR (%). All four types share each model's original positive-correct set, whose eligibility is reported in Table A9. Negative refusals remain in the denominator; source precision is preserved.

MLLM	Obj.	Assoc.	Attr.	Rel.
Qwen3-VL	1.7	7.8	12.1	32.3
Qwen2.5-VL	3.3	17.0	21.7	43.4
InternVL3.5	6.7	15.3	20.0	28.7
LLaVA-OV	66.7	69.1	81.5	87.7
LENS	23.4	18.8	31.2	36.7
Orsta	0.0	10.1	11.5	21.6
Seg-R1	3.9	19.0	23.9	49.3
Seg-Zero	5.6	20.3	23.7	47.4
TreeVGR	0.0	7.1	6.3	13.5
UniVG-R1	25.91	34.41	47.37	55.06
Vision-R1	0.0	8.1	8.1	14.9
VisionReasoner	5.3	29.5	30.9	55.1
Visual-RFT	17.39	24.64	35.27	49.76

Table A7: Joint-grounding CBR (%). All four types share each model's original positive-correct set, whose eligibility is reported in Table A9. Negative refusals remain in the denominator; source precision is preserved.

MLLM	Direct Eligible (%)	Joint Eligible (%)	Direct mIoU	Joint mIoU
Qwen3-VL	47.6	46.4	0.4648	0.4585
Qwen2.5-VL	48.4	42.4	0.4672	0.4223
InternVL3.5	46.8	30.0	0.4653	0.3193
LLaVA-OV	37.2	16.2	0.3351	0.2201
LENS	49.6	25.6	0.4874	0.2520
Orsta	30.2	27.8	0.3169	0.2948
Seg-R1	52.2	41.0	0.4976	0.4059
Seg-Zero	55.0	46.4	0.5222	0.4549
TreeVGR	26.2	25.2	0.2811	0.2749
UniVG-R1	50.8	49.4	0.4936	0.4820
Vision-R1	29.6	29.6	0.3098	0.3046
VisionReasoner	50.6	41.4	0.4959	0.4167
Visual-RFT	42.2	41.4	0.4114	0.4249

Table A9: Original positive-correct coverage (Eligible, %) and all-positive mIoU under both grounding protocols. Description results appear in Table A4. Both protocols cover all 13 models.

sults. LENS lowers FGR from 94.20% to 28.55%, while positive success falls from 49.6% to 25.6% and mIoU from 0.4874 to 0.2520. Seg-Zero lowers FGR from 97.10% to 38.00%, with positive success 55.0% to 46.4% and mIoU 0.5222 to 0.4549. LLaVA-OV instead increases FGR from 33.65% to 93.20%, while positive success falls from 37.2% to 16.2%; its joint relation CBR is 87.7%. Across all 13 models, joint FGR

MLLM	Direct FGR	Joint FGR	Desc. err.
Qwen3-VL	21.15	22.45	9.20
Qwen2.5-VL	34.50	33.85	4.00
InternVL3.5	61.25	39.85	6.40
LLaVA-OV	33.65	93.20	7.30
LENS	94.20	28.55	55.60
Orsta	44.15	37.55	7.05
Seg-R1	61.75	31.60	5.90
Seg-Zero	97.10	38.00	14.55
TreeVGR	75.60	59.50	0.70
UniVG-R1	99.80	87.40	0.45
Vision-R1	37.80	43.80	5.70
VisionReasoner	80.85	45.95	12.55
Visual-RFT	28.25	65.55	1.50

Table A8: Unfiltered region emission and joint-description error (%). All three rates use 4,000 negative queries per model. Description error scores the generated joint-task text; FGR counts parsed coordinate emissions. Positive grounding quality appears in Table A9.

averages 48.25%, positive success 35.60%, and mIoU 0.3639. UniVG-R1 and Visual-RFT have joint FGR 87.40/65.55%, success 49.4/41.4%, and mIoU 0.4820/0.4249. Protocol changes therefore affect both rejection and localization.

Joint-description error uses 4,000 negative queries per model. Qwen3-VL, UniVG-R1, and Visual-RFT have error rates of 9.20%, 0.45%, and 1.50%. LENS has 55.60%, versus 4.20% object error in query-free captions. These measures have different tasks and denominators, so the percentages should not be interpreted as a shared error event.

C.8 Language–Region Disagreement

Differences between aggregate verification and grounding rates must be distinguished from disagreements on matched queries. The aggregate positive rate gap is

$$G = 100 \max\{0, \mathrm{FGR}_{\mathrm{DG}} - \mathrm{FA}_{\mathrm{verify}}\}. \quad (8)$$

Here rates are fractions, and verification FAR uses valid judgments. The gap is 35.00 points for InternVL3.5 ($61.25 - 26.25$) and 56.88 for LENS ($94.20 - 37.32$). Different denominators and unavailable joint events prevent interpreting this difference as a query-matched mismatch rate.

Query-matched responses. Matching by model and query identifier, 26.72% of matched negative-query records across 12 models combine an explicit unsupported verification judgment with a direct-grounding emission. The main text reports a 22.60–28.47% range across the four edit types. This is evidence about independent responses to the same query. This analysis is restricted to its 12-model matched subset and is kept distinct from the complete 13-model tables; its rate is not extrapolated to unmatched records.

D Verifier Training and Mitigation Analysis

We evaluate shared candidate support and structural comparison under a fixed-candidate protocol, reporting both their rejection gains and positive-grounding costs. The ASV settings, ablations, and mitigation results use the same evaluated candidates as the main paper. Appendix D.11 separately documents an earlier GRPO training study.

D.1 Evaluation Protocol

Unfiltered predictions, shared support, and full Adaptive Structured Verification (ASV) are compared on the same 13 upstream MLLMs. Each policy fixes the upstream candidate boxes, the original positive reference predictions, and the originally eligible positive groups. A retained prediction preserves its emission state and coordinates; a rejected prediction becomes empty. Empty upstream predictions remain empty. Consequently, mitigation evaluates rejection of existing assignments rather than proposal generation or coordinate refinement.

Each model retains 1,000 positive and 4,000 negative queries, including 1,000 negatives per edit type. FGR counts parsed coordinate emissions after model-specific coordinate conversion, using the full negative-query denominator. CBR fixes each model's original positive-correct groups and positive boxes; negative refusals remain in its denominator. All-positive mIoU assigns zero to empty, rejected, or invalid positive predictions. Aggregate results weight models equally, including the mean of per-model CBR ratios. Table 3 gives the main operating points; the tables below retain this evaluation contract.

D.2 Verifier Adaptation and Calibration

JEV-2B adapts the TypeSafe/Open-Jev scoring interface on Qwen3.5-2B (Almeida, 2026; Cai, 2026; Qwen Team, 2026). Frozen OmniVerifier-7B provides a second, image-conditioned score (Zhang et al., 2026); all upstream MLLMs also remain frozen. JEV reads the final prompt representation with a scalar head, whereas OmniVerifier uses the true-minus-false token-logit difference. Thus the shared-support input combines two readout interfaces for the same image, expression, and candidate region.

JEV initializes new rank-8 LoRA adapters with scale 16 on Qwen3.5-2B (Hu et al., 2022). Adapters cover both full-attention projections (`q/k/v/o_proj`) and linear-attention projections (`in_proj_qkv`, `out_proj`). A float32 scalar head reads the final 2,048-dimensional prompt state. Its training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{NLL}} + 0.1\mathcal{L}_{\mathrm{Brier}}, \tag{9}$$

where the Brier component penalizes squared probability error (Brier, 1950). The adaptation dataset contains 2,000 COCO images and 16,000 candidate–expression records, balanced between KEEP and REJECT. The saved training procedure shuffles records with seed 63, reserves the first 512 for temperature fitting, and trains on the remaining 15,488. Training uses 1,000 optimizer updates, batch size 4, gradient accumulation 4, an adapter learning rate of 5×10^{-5}, and a head learning rate of 10^{-4} on one GPU. AdamW uses zero weight decay; gradient norms are clipped at 1.0. LoRA contributes 2,506,752 trainable parameters and the scalar head adds 2,049, totaling 2,508,801 parameters (approximately 0.113% of the 2.22B verifier).

Optimization dynamics. Figure A4 plots all 40 logged NLL and Brier points, each averaged over 25 optimizer updates. Mean NLL decreases from 0.3239 over the first 100 updates to 0.1355 over the last 100; mean Brier decreases from 0.1002 to 0.0317. The run processes 16,000 record presentations, approximately 1.033 passes over the training split. The fixed-order loop completes its first pass at update 968; the final 25-update window therefore contains revisited training records. The traces characterize optimization on the training data; no checkpoint-wise, image-disjoint validation curve was recorded. The plotted values are included in the source supplement.

Temperature fitting and data separation. The 512 calibration records are excluded from gradient updates, but their 461 images also occur in training records. This is a record-level holdout rather than an image-disjoint validation set. Temperature scaling (Guo et al., 2017) selects $T = 1.7766$ and changes expected calibration error from 0.0305 to 0.0114 and negative log-likelihood from 0.1525 to 0.1197 on this temperature-fitting set. These values are fitting diagnostics. A positive temperature preserves scalar-score ordering. The training–benchmark image intersection has not been audited against the final 1,000-image benchmark manifest; confirming this separation requires the

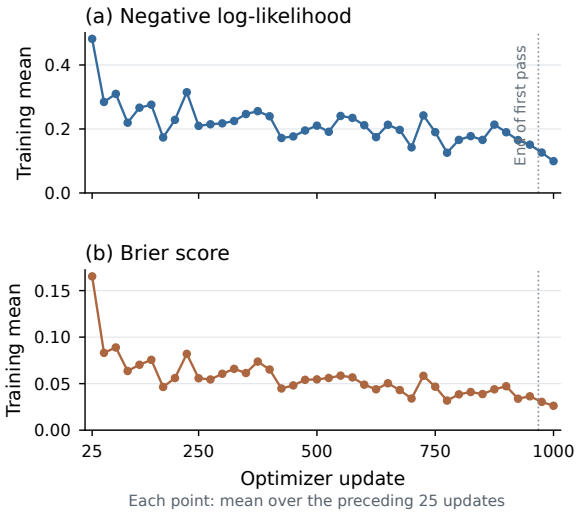


Figure A4: Training dynamics of the scalar-head JEV used by ASV. Each of the 40 points averages the preceding 25 optimizer updates; values retain the logged precision. The dotted line marks completion of the first data pass at update 968. These are training-set measurements.

corresponding image identifiers. Benchmark acceptance heads and thresholds use the cross-fitting protocol below.

D.3 Decision Heads and Predicate Routing

For direct grounding, feature order is $(s_{\mathrm{O},0}, s_{\mathrm{J},0})$ for the support head f_0 and $(s_{\mathrm{O},0}, s_{\mathrm{J},0}, \delta_{\mathrm{O}}, \delta_{\mathrm{J}})$ for the structural head f_1. Each upstream model and evaluation fold has separate logistic heads. Fitting uses positive/negative expression labels on usable rows; only threshold selection restricts positives to originally correct candidates. Features use training-fold means and standard deviations, with unit scale for constant features. The unweighted binary logistic objective uses an L_2 coefficient of 1.0 on feature weights and an unpenalized intercept. Optimization starts at zero and uses up to 300 Newton updates, stopping when the largest damped update is below 10^{-9}.

Routing longest-matches the lowercased query against a frozen predicate inventory. Structural comparison applies to *behind*, *in front of*, *under*, *on top of*, *above*, and *below*, provided both original scores, both alternative-score maxima, and a fitted structural head are available. Replacement expressions preserve the object mentions, exclude the original expression, and are deduplicated. Outside this scope, shared support supplies the final decision. Inside it, the final decision is the conjunction of support and structural decisions.

Each head requires at least 30 usable fitting rows and both labels. An unavailable support head or missing original scores preserves an emitted upstream candidate. An unavailable structural head, unmatched opposed predicate, or missing competitive scores falls back to shared support. These rules preserve $k_i \le k_i^{(0)}$: the full keep set is a subset of the support keep set, itself a subset of the upstream candidates. With fixed eligibility and positive reference boxes, FGR and CBR therefore cannot increase relative to shared support. This subset property alone does not establish a better operating point: additional rejection may also remove correct positives, so retention and all-positive mIoU must be reported jointly.

D.4 Cross-Fitted Threshold Selection

Direct-grounding evaluation uses image-level two-fold cross-fitting, assigning membership by MD5(image basename) mod 2. All paired queries from an image stay together. For each evaluation image, acceptance heads and thresholds are fitted on the other fold; head fitting and threshold selection share that training fold. The common nominal retention target is 0.95 on originally correct positives.

Let N_c denote originally correct calibration positives, U unrouted correct positives retained by support, and V_0 routed correct positives also retained by support. For $N_c > 0$, fixing the support decision constrains the retention attainable by changing η_1 to

$$\frac{U}{N_c} \le \widehat{R}_{\mathcal{C}}(\eta_1) \le \frac{U + V_0}{N_c}. \tag{10}$$

Finite samples and tied scores further discretize attainable values. The implementation selects η_0 by an order statistic of eligible training-fold positive support scores. To select η_1, it subtracts retained unrouted positives from the desired total, clips the required routed count to the number of covered positives, and takes the corresponding descending structural-score order statistic. A zero required count sets $\eta_1 = +\infty$; ties are retained through $\ge$. This structural threshold is selected before conjunction with $k_i^{(0)}$, whereas achieved retention is measured after the final conjunction. Nominal and achieved retention therefore need not coincide.

D.5 Support and Structural Ablations

Table A10 reports the available stage ablation on the same 13 models. Shared support lowers mean FGR from 59.23% to 22.55%; adding structural

comparison lowers it to 19.16%. These are reductions of 36.68 and 3.39 percentage points, respectively, totaling 40.07 points (67.65% relative to the unfiltered mean). Correct-grounding retention is 94.26% for support and 91.71% for full ASV. All-positive mIoU changes from 0.4268 unfiltered to 0.4011 with support and 0.3910 with ASV. The structural stage thus adds rejection at a cost of 2.55 retention points and 0.0101 mIoU.

Measure	Before	Support	Full	Δ
FGR	59.23	22.55	19.16	-3.39
EOH FGR	48.11	12.20	11.94	-0.26
ROH FGR	70.36	32.89	26.38	-6.51
Object CBR	16.47	1.69	1.69	+0.00
Assoc. CBR	25.63	9.22	9.07	-0.15
Attribute CBR	33.96	8.32	8.19	-0.13
Relation CBR	42.79	27.14	19.34	-7.80
R_{correct}	100.00	94.26	91.71	-2.55

Table A10: Equal-model means for the 13-model mitigation evaluation. The final column is full minus shared support in percentage points. EOH/ROH are FGR strata; the four individual edit columns use CBR.

The largest incremental changes concern relational errors: ROH FGR falls from 32.89% to 26.38% (6.51 points), compared with EOH FGR from 12.20% to 11.94% (0.26 points). Relation CBR falls from 27.14% to 19.34% (7.80 points); association and attribute CBR fall by 0.15 and 0.13 points, while object CBR remains 1.69% at reported precision. Both semantic-level FGR means and all four CBR means improve relative to unfiltered predictions.

Figure A5 summarizes the same stage-wise CBR changes. Read with the unfiltered profiles in Figure A3, it shows where rejection concentrates: shared support reduces all four error types, while the added structural stage mainly reduces relation reuse.

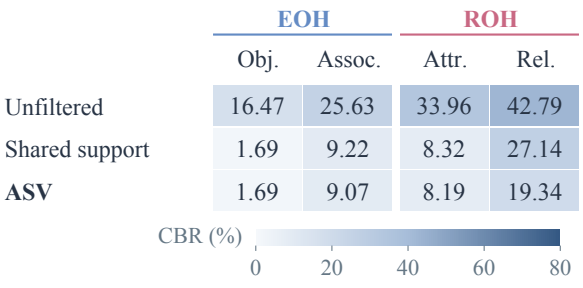


Figure A5: Four-type CBR through the two mitigation stages. Cells show equal-model means over 13 MLLMs with fixed candidates and eligibility. The shared 0–80% scale also applies to Figures A3 and A6. Retention costs appear in Table A10.

Ablation scope. This comparison isolates the additional structural stage conditional on the existing two-verifier support policy. OmniVerifier-only and JEV-only ablations, an unadapted-JEV baseline, and separate removal of each structural contrast are not reported. It therefore does not quantify either verifier's independent contribution or establish that both contrasts are individually necessary. Because the full policy can only reject more candidates, the matched-retention comparison below is needed to assess whether its incremental relation gains extend beyond a lower-retention operating point.

D.6 Matched-Retention Analysis

The matched-retention analysis covers 12 MLLMs. Both policies use the same 11 retention targets spanning 0.80–0.99. For each model, shared support at target 0.95 defines the achieved-retention anchor; the full policy is linearly interpolated at that anchor within its observed curve range. Table A11 reports equal-model means and paired model-bootstrap intervals from 10,000 resamples. Both relation metrics improve at the matched anchors.

Metric	Support	Full	Δ [95% CI]
Rel. FGR	47.95	40.41	$-7.54\,[-9.03, -5.99]$
Rel. CBR	28.76	23.18	$-5.58\,[-7.05, -4.05]$

Table A11: Matched-retention relation results across 12 MLLMs. Rates are percentages; differences and intervals are percentage points. Full-policy rates use linear interpolation. Intervals resample paired differences across upstream models.

This analysis has a narrower scope than the 13-model nominal-target comparison. The excluded model and its exclusion reason are unspecified, as are the 11 target values. These omissions prevent exact reconstruction of the matched subset and sweep from the text alone. The intervals describe paired variation across the reported models; they are not image-bootstrap intervals or uncertainty over retraining the verifiers. Interpolated operating points are also distinct from directly observed threshold settings.

D.6.1 Serial Threshold Calibration

The serial sensitivity probe preserves the all-eligible-positive target count but selects the structural-score order statistic only among support-retained routed positives, using the same image folds and final conjunction. Across the same 13

models, it attains FGR 20.13%, versus 22.55% for shared support, with correct retention 92.49% versus 94.26%. This sensitivity check of threshold construction still differs from shared support in achieved retention and does not constitute an additional matched-retention test.

D.7 Evidence Probes and Computational Cost

Table A12 specifies the image-swap and predicate-centering probes. Image swaps produce a mean OmniVerifier logit shift of 11.41 across 13 MLLMs, indicating sensitivity to image evidence with the query and candidate coordinates held fixed. On probe-covered candidates, subtracting the model-specific predicate mean from OmniVerifier's negative contrast $-\delta_{\mathrm{O}}$ yields mean expression-validity AUROC 0.598, compared with 0.613 before centering. Mean predicate-identity R^2 is 0.080 across 13 models. The centered signal retains discrimination beyond a predicate-mean prior; this diagnostic is not itself a mitigation ablation or a causal explanation of a particular decision.

Probe	Protocol and aggregation
Image swap	Keep the query and candidate box coordinates fixed; score the original and mismatched images. Average paired logit shifts within each model, then across 13 models.
Predicate centering	Subtract each model's predicate mean from $-\delta_{\mathrm{O}}$ on covered candidates. Use expression-validity labels for AUROC; average per-model AUROC and the predicate-explained variance fraction R^2.

Table A12: Evidence-probe protocols. Each upstream model receives equal weight in the aggregate probe statistics.

For a non-empty candidate, two original scores and K alternatives per verifier require $2 + 2K$ scorer–expression evaluations; support alone requires two. The CPU policy evaluation consumes previously collected scores. The recorded predicate-and-score-availability audit activates structural evidence in 20.97% of cases; additional OmniVerifier rival scores amount to 41.94% of its original support-score evaluation budget. These quantities count scorer–expression evaluations, not wall-clock latency or total GPU cost. The OmniVerifier-specific overhead does not measure the complete two-verifier system overhead.

D.8 Per-Model Mitigation Results

Tables A13 and A14 retain all per-model results behind the 13-model means. Every model has lower FGR with full ASV than with shared support. Per-model CBR and retention must nevertheless be read together: their magnitudes vary with each model's original candidates and eligible positives.

With structural comparison, Seg-Zero's relation CBR changes from 35.64% to 25.09%, with retention from 93.82% to 91.64%; LENS changes from 40.73% to 27.02% and from 93.95% to 91.94%, respectively. TreeVGR changes from 7.63% to 6.11% relation CBR and from 94.66% to 90.08% retention. Vision-R1 changes from 8.78% to 8.11% and from 93.24% to 91.89%. Qwen2.5-VL changes from 32.64% to 21.07% and from 95.04% to 90.91%. Figure A6 uses a common 0–80% CBR scale for TreeVGR, Vision-R1, and Qwen2.5-VL; the LENS and Seg-Zero profiles appear in Figure 6.

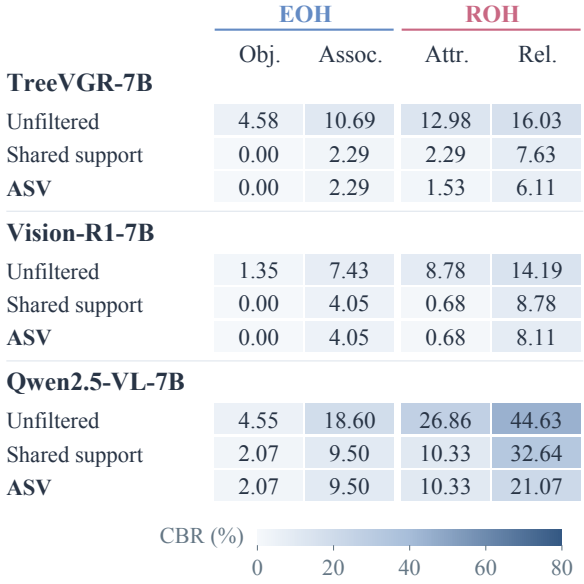


Figure A6: Three-policy CBR heatmaps for TreeVGR, Vision-R1, and Qwen2.5-VL, using fixed candidates and eligible positives with a common 0–80% scale. Tables A13 and A14 report the accompanying retention and all-positive mIoU.

D.9 Joint-Grounding Extension

The joint extension evaluates all 13 models using a separately fitted configuration and scores collected on their joint-grounding candidates. UniVG-R1 and Visual-RFT normalized coordinates are converted to image pixels before scoring. The support head uses the OmniVerifier original score and

MLLM	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL	14.00	6.60	21.40	1.26	7.98	8.40	24.37	94.54	0.4379
Qwen2.5-VL	20.50	11.10	29.90	2.07	9.50	10.33	32.64	95.04	0.4435
InternVL3.5	27.55	14.90	40.20	2.56	12.82	13.68	30.77	94.87	0.4420
LLaVA-OV	20.10	10.00	30.20	2.69	10.75	7.53	30.11	93.55	0.3131
LENS	27.80	16.00	39.60	3.23	16.13	13.31	40.73	93.95	0.4562
Orsta	20.65	11.30	30.00	0.00	2.65	2.65	11.26	94.04	0.2936
Seg-R1	24.25	13.40	35.10	2.68	10.73	11.11	35.63	95.40	0.4728
Seg-Zero	26.30	14.80	37.80	2.55	12.36	11.64	35.64	93.82	0.4868
TreeVGR	25.20	13.80	36.60	0.00	2.29	2.29	7.63	94.66	0.2652
UniVG-R1	24.95	13.60	36.30	1.57	9.84	8.66	28.74	94.49	0.4642
Vision-R1	21.05	11.20	30.90	0.00	4.05	0.68	8.78	93.24	0.2906
VisionReasoner	25.40	13.80	37.00	2.37	12.25	12.65	37.55	94.47	0.4663
Visual-RFT	15.35	8.10	22.60	0.95	8.53	5.21	28.91	93.36	0.3820

Table A13: Shared support verification: per-model results corresponding to the 13-model means in Table 3. EOH/ROH are FGR strata and the four edit columns report CBR. Rates are percentages; mIoU is in 0–1. The eligible positive set is fixed within each model.

MLLM	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL	12.55	6.60	18.50	1.26	7.98	8.40	18.49	92.02	0.4265
Qwen2.5-VL	17.35	11.00	23.70	2.07	9.50	10.33	21.07	90.91	0.4248
InternVL3.5	23.35	14.80	31.90	2.56	12.82	13.68	19.66	91.88	0.4281
LLaVA-OV	16.75	9.90	23.60	2.69	10.75	7.53	21.51	90.32	0.3032
LENS	23.15	15.80	30.50	3.23	15.73	13.31	27.02	91.94	0.4461
Orsta	16.90	11.20	22.60	0.00	2.65	2.65	7.28	90.73	0.2834
Seg-R1	20.25	12.90	27.60	2.68	10.34	11.11	23.37	92.72	0.4596
Seg-Zero	22.40	14.40	30.40	2.55	12.36	11.64	25.09	91.64	0.4754
TreeVGR	19.45	12.90	26.00	0.00	2.29	1.53	6.11	90.08	0.2551
UniVG-R1	22.85	12.80	32.90	1.57	9.06	8.27	24.02	93.70	0.4607
Vision-R1	18.05	11.20	24.90	0.00	4.05	0.68	8.11	91.89	0.2874
VisionReasoner	22.05	13.80	30.30	2.37	12.25	12.65	26.48	92.49	0.4564
Visual-RFT	13.95	7.90	20.00	0.95	8.06	4.74	23.22	91.94	0.3761

Table A14: Adaptive Structured Verification: per-model results corresponding to the 13-model means in Table 3. EOH/ROH are FGR strata and the four edit columns report CBR. Rates are percentages; mIoU is in 0–1. The eligible positive set is fixed within each model.

the stored raw JEV scalar-head logit. The structural head adds only the OmniVerifier alternative-minus-original margin: three features for joint grounding, versus four for direct grounding. Both configurations use monotone conjunction and preserve the original box coordinates.

Joint head fitting uses candidate-correctness labels: a positive query receives label one only when upstream IoU is at least 0.5. Routing checks whether a collected rival surface belongs to the opposed-predicate set. Two-fold membership is determined by MD5 of the base image-group identifier, keeping paired queries together. Fitting and threshold selection use the opposite fold from evaluation; feature standardization uses only the fitting fold. Table A15 makes the task-specific differences explicit. These separate task configurations preclude interpreting the joint experiment as a controlled ablation of the fourth direct-grounding feature.

Task	Head-fitting label $y = 1$	Structural features
Direct	Positive expression	Two original scores and both verifier contrasts
Joint	Positive query with upstream IoU ≥ 0.5	Two original scores and OmniVerifier contrast

Table A15: Task-specific decision-head configurations. Other fitting rows have $y = 0$; retention thresholds use originally correct positives. Both tasks use image-grouped two-fold cross-fitted acceptance heads.

At nominal target 0.95, joint eligibility fixes each model's original positives with IoU ≥ 0.5. The full 1,000-positive and 4,000-negative denominators, unchanged positive reference boxes, and retained negative refusals follow the contract above. Equal-model FGR is 48.25% unfiltered, 19.70% with shared support, and 18.45% with the full extension. Full ASV reduces FGR in all 13

models relative to both unfiltered predictions and shared support. All four mean CBR values decrease relative to unfiltered predictions; no per-model CBR increases. Table A16 reports mean CBR and positive-grounding outcomes, while Table A17 supplies the underlying per-model operating points.

Metric	Unfiltered	Support	ASV
FGR	48.25	19.70	18.45
Object CBR	12.30	1.54	1.54
Association CBR	21.62	9.01	8.79
Attribute CBR	27.20	8.43	7.75
Relation CBR	41.18	25.18	23.08
Correct retention	100.00	93.95	92.99
mIoU	0.3639	0.3379	0.3341

Table A16: Joint-grounding mitigation: equal-model means over 13 models at nominal target 0.95. Eligibility and positive reference boxes are fixed. Rates are percentages; mIoU includes all positives.

The joint extension retains 92.99% of originally correct positives, compared with 93.95% for support; all-positive mIoU is 0.3341, compared with 0.3379 for support and 0.3639 unfiltered. Relation CBR decreases from 25.18% with support to 23.08% with full ASV. As in direct grounding, the reduction in false assignments accompanies a measurable positive-grounding cost.

D.10 Qualitative Attention Analysis

Figure A7 reserves space for cases to be selected in a later revision. No attention maps or case-specific conclusions are supplied here. The intended comparison pairs recorded four-task outputs with model attention and verifier evidence while retaining the recorded ASV keep/reject decisions. Any auxiliary inspection prompt will be identified with its panel. In particular, the Direct-*G* response in Figure 7 uses an additional reasoning-and-answer instruction; it is distinct from the standard evaluation interface in Appendix C.1.

The existing extraction protocol uses teacher-forced replay of saved MLLM responses, averaging image-patch attention across the last four decoder layers, all heads, and selected decision, entity, or coordinate tokens. OmniVerifier maps use the final prompt token producing support logits. Verifier probes use each grounding candidate, the Direct-*G* candidate for its matched verification query, and the full image for the captioned claim. Each map is divided by its patch mean; each pair shares a pooled 95th-percentile upper color limit and a 50% overlay. These conventions will be held fixed across the selected cases. Attention visualizations are descriptive diagnostics and do not by themselves establish causal explanations of model decisions (Jain and Wallace, 2019).

Attention-case placeholder

Cases to be selected in a later revision.
Four-task outputs, model attention, verifier evidence, and recorded ASV decisions.

Figure A7: Reserved qualitative comparison. Case selection and attention panels are pending; this placeholder makes no empirical claim.

D.11 Exploratory GRPO Verifier Training

We additionally document an earlier generative-verifier experiment using GRPO (Shao et al., 2024) in ms-swift (Zhao et al., 2025). Two Qwen3.5-2B runs, R1 and R2, generate `binding` and `decision` fields from the same training set of 2,000 images and 16,000 candidate–expression records, balanced between KEEP and REJECT. These runs are separate from the scalar-head JEV adaptation in Appendix D.2; they do not produce the ASV results reported in the main tables. Table A18 records their saved configurations. This supplement reports the optimization dynamics of the generative verifier.

Reward configurations. R1 combines format, ROH binding, and decision components as $R_1 = 0.1r_F + r_B + r_D$. Its component values are $r_F = 0.1$ for a valid typed response and zero otherwise; $r_B = 1.5$ for a correct ROH binding and zero otherwise; and $r_D = 1$ for a correct decision, -0.2 for an unparseable decision, and zero for a parsed incorrect decision. Binding targets are derived from the ROH type and KEEP/REJECT label, not independently annotated process supervision.

R2 uses $R_2 = 0.1r'_F + r'_D$, omitting binding reward. Here $r'_F = 0.1$ for valid format and -0.2 otherwise; its decision component is evaluated from the parsed decision field. Both configurations therefore give valid format an effective contribution of 0.01. Their parsers and invalid-output penalties also differ, so R1 versus R2 is a reward-configuration comparison rather than an isolated binding-reward ablation.

Optimization dynamics. Figure A8 uses all 400 recorded steps from each run. Every reward and KL value agrees with the corresponding checkpoint-400 training history. Rewards are

Shared support verification										
MLLM	Eligible (%)	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL	46.4	14.50	7.30	21.70	1.29	6.47	6.90	22.84	92.24	0.4220
Qwen2.5-VL	42.4	17.85	9.50	26.20	2.36	9.43	11.32	27.83	93.87	0.3913
InternVL3.5	30.0	21.25	11.80	30.70	1.33	8.67	7.33	18.00	94.67	0.2966
LLaVA-OV	16.2	23.10	12.70	33.50	1.23	17.28	14.81	43.21	96.30	0.2056
LENS	25.6	9.60	4.90	14.30	2.34	7.03	4.69	25.00	91.41	0.2269
Orsta	27.8	16.65	8.70	24.60	0.00	6.47	5.04	12.23	95.68	0.2717
Seg-R1	41.0	17.50	9.00	26.00	1.95	10.73	11.22	33.66	93.66	0.3766
Seg-Zero	46.4	18.60	9.30	27.90	2.16	9.05	8.62	30.17	94.40	0.4245
TreeVGR	25.2	22.20	11.80	32.60	0.00	3.17	3.17	6.35	94.44	0.2539
UniVG-R1	49.4	26.95	15.30	38.60	2.02	11.34	10.53	32.39	92.71	0.4472
Vision-R1	29.6	21.45	11.20	31.70	0.00	3.38	1.35	6.08	92.57	0.2830
VisionReasoner	41.4	23.15	12.80	33.50	2.90	14.49	13.04	39.61	95.65	0.3950
Visual-RFT	41.4	23.25	12.50	34.00	2.42	9.66	11.59	29.95	93.72	0.3986
ASV: joint-grounding extension										
MLLM	Eligible (%)	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL	46.4	13.30	6.80	19.80	1.29	6.47	6.47	20.26	90.09	0.4122
Qwen2.5-VL	42.4	16.40	9.00	23.80	2.36	9.43	10.38	24.06	93.40	0.3893
InternVL3.5	30.0	20.10	11.50	28.70	1.33	8.67	6.67	16.00	94.67	0.2960
LLaVA-OV	16.2	20.30	10.90	29.70	1.23	14.81	12.35	38.27	92.59	0.1972
LENS	25.6	9.25	4.90	13.60	2.34	7.03	4.69	23.44	91.41	0.2269
Orsta	27.8	14.85	8.10	21.60	0.00	6.47	3.60	12.23	92.81	0.2598
Seg-R1	41.0	17.10	8.90	25.30	1.95	10.73	11.22	32.20	93.66	0.3766
Seg-Zero	46.4	18.20	9.30	27.10	2.16	9.05	8.62	28.02	94.40	0.4244
TreeVGR	25.2	21.30	11.50	31.10	0.00	3.17	2.38	6.35	94.44	0.2532
UniVG-R1	49.4	26.35	15.20	37.50	2.02	10.93	10.53	30.36	92.31	0.4450
Vision-R1	29.6	18.80	10.20	27.40	0.00	3.38	0.68	6.08	91.22	0.2766
VisionReasoner	41.4	21.95	12.30	31.60	2.90	14.49	12.08	36.23	94.20	0.3894
Visual-RFT	41.4	21.95	12.00	31.90	2.42	9.66	11.11	26.57	93.72	0.3968

Table A17: Per-model joint-grounding extension results. The four edit columns report CBR; Eligible gives the original positive-correct proportion (%), fixed across policies and types. FGR counts parsed coordinate emissions after model-specific coordinate conversion, as specified in Appendix D.9. Rates are percentages; mIoU uses all positive queries.

also reproduced by weighting the logged component means with the saved reward weights. Both runs' smoothed rewards increase from their initial values; final logged KL is 0.02337 for R1 and 0.01336 for R2. Different reward scales preclude comparing absolute reward heights as model quality, and training curves alone do not establish held-out grounding performance.

Training diagnostics. The saved configuration uses eight GPUs: per-device batch 1 and accumulation 8 yield 64 completions per update, corresponding to eight prompt records with eight samples each. Both runs end at epoch 0.2 after 400 updates. No logged completion is length-clipped; the gradient norm exceeds the configured clip threshold of 1.0 in 4/400 R1 steps and 6/400 R2 steps. These observations and the bounded recorded KL support numerical stability over the observed run, but do not establish convergence. Evaluation is disabled and no validation split or validation-based checkpoint selection is recorded; generalization therefore remains unmeasured by these training traces.

Setting	R1 and R2
Base model	Qwen3.5-2B, revision `15852e8c`
Training data	2,000 images; 16,000 records (8,000 KEEP / 8,000 REJECT)
LoRA	Rank 32, scale 32, dropout 0.05
Target projections	`q_proj`, `k_proj`, `v_proj`, `o_proj`
Optimizer	Fused AdamW; LR 2×10^{-6}; weight decay 0.1
Adam parameters	$(\beta_1, \beta_2) = (0.9, 0.95)$; $\epsilon = 10^{-8}$
Schedule / updates	Cosine; warmup ratio 0.05; 400 steps
GPU count / per-device batch / accumulation	8 / 1 / 8
Generation batch / group size	64 completions / 8 per prompt
Effective update / data exposure	64 completions; 8 prompt records per update; 0.2 epoch
Sampling	Temperature 1.0; at most 64 new tokens; thinking disabled
GRPO controls	KL coefficient 0.01; clipping $\epsilon = 0.2$; group reward scaling
Precision / gradient controls	bfloat16; checkpointing enabled; norm clip 1.0
Randomness / data split	Model and data seeds 63; no automatic evaluation split

Table A18: Saved settings shared by the two historical GRPO runs. The generation batch is completion-level. The 400 updates correspond to 3,200 record presentations, not 25,600 distinct prompts. Both runs use the same historical verifier training file; these settings do not describe the JEV adaptation pool.

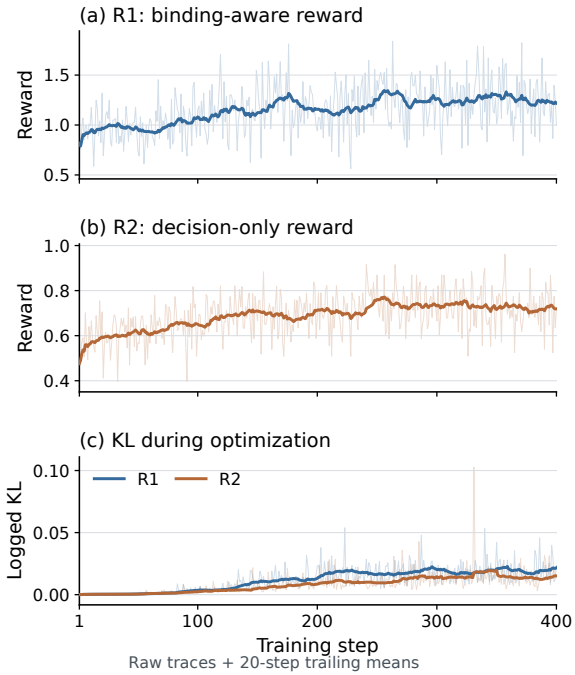


Figure A8: GRPO optimization over 400 steps. Faint traces retain every step; solid curves show trailing 20-step means. Reward panels use separate scales.